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### **Analyzing Online Commerce Trends via Similarity Metrics**

E-commerce and reselling businesses exist within a vast, competitive landscape. Since the emergence of new technologies, people have been utilizing every tool at their disposal to gain an edge in business performance and sales. There are a multitude of methods for analyzing business insights, but one of the most important, if not the most important, is understanding the customer and what they desire. But what exactly does that mean? How do we determine what the customer wants and where their dollar is likely to go? Ultimately, different methods and reasoning exist, and similarity metrics can be utilized to address these questions. The specific stakeholder in question that can benefit from such analysis is an e-commerce business owner who seeks to resell online products and make a profit. By utilizing similarity metrics, they can discover and analyze customer bases and their buying patterns, allowing them to cater to customers within a given market more effectively. For example, if one wants to resell chargers, they can analyze what chargers are frequently being bought and what other products those users are buying (as one example of a similarity metric). Other methods exist to help one get to the bottom of the situation as well.

The data utilized to address this stakeholder question is an Amazon dataset from Kaggle that has nearly 5000 data entries and displays a list of technology products and accessories sold within a given time period. This data was comprehensive and well put together, but still needed some cleaning in order to be ready for insights. For example, the currency symbol had to be taken out of the price and discounted price in order for the code to function and process the numbers to compare products based on price. Additionally, the category names were properly separated in order to effectively be read and interpreted by the Python code. Repeating names were also frequent in the data, as different sellers were selling the same product, so that also had to be taken into account.

When it comes to comparing products with different attributes cosine similarity is a particularly useful metric for analyzing high-dimensional data. A systematic procedure is followed by cosine similarity to analyze the product data. Originally a single text field was used to house text features like `about_product` `review_title` and `review_content`. Then using `CountVectorizer` this combined text was vectorized into a high-dimensional vector space where each dimension represents a distinct word or term. Furthermore standardization was applied to

numerical features such as `rating_count` and `discount_percentage` to guarantee that they were on a similar scale. In this high-dimensional space each product was represented as a vector with elements denoting the standardised numerical attributes and the presence or magnitude of particular words (text features). Instead of focusing on the magnitude of two vectors cosine similarity measures the cosine of their angle which reflects their directional alignment. This computation makes use of the cosine similarity formula. The analysis finds products with similar text descriptions reviews and numerical features by calculating the cosine similarity between product vectors. The most similar products are shown by sorting these similarity scores which makes it easier to make recommendations and group related products together.

As can be observed, The most similar products are other chargers and wires. The brand most commonly observed on the list is Wayona, which demonstrates users' desire to buy products from this brand. Most products are similar in name and nature, with differences, such as length, number of products in a pack, and the specific cable type (e.g. USB, lightning, or USB-C). The cosine similarity was done measuring all metrics aside from `'product_name'`, `'product_id'`, `'discount_price'`, and `'actual_price'`. Such was done to provide a more useful analysis on the types of products that were similar to the original product that was being compared in the Python file. Thus, if a stakeholder aims to find a similar product to sell, they are provided with useful alternatives, as opposed to the same product name repeated throughout the lines. By taking factors like ratings, user id, and number of reviews into account, this can be a useful tool for a stakeholder to use in order to find similar products being consumed by a similar user base.

Although drawbacks do exist, cosine similarity provides valuable insights for e-commerce. Vector representation is the core of cosine similarity, meaning data quality is of paramount importance. Inaccurate similarity scores can result from poor handling of text features, such as neglecting to stem words or removing stop words. After all, cosine similarity ignores magnitude and places its focus on the angle of the two relevant vectors. This causes factors like significant cost or rating count to potentially be an issue, as it can cause other important product factors not to be considered much. The pairwise comparisons increasing for larger datasets can spell trouble for even more extensive data, causing there to be a reconsideration of results.. Synonyms and specific category words may also mislead, as the

python script cannot fully comprehend what the context of what the data is mentioning. Odd normalization metrics can also taint results, producing outcomes that are odd themselves. Cosine similarity is one form of similarity, after all, and has its clear limits; this is not to be of surprise. Yet with these limitations, it can still provide useful data to interpret and be used by a stakeholder.

Upon examining this data with Euclidean similarity, various insights were made, considering the primary metrics within the csv file, and excluding product\_name product\_id actual\_price and discounted\_price, to find useful and non-repetitive results. Similar products were sorted by factors such as discount rates, rating counts, and the similarity in ratings, showing which products customers prefer and gravitate toward. Certain companies, such as Amazon Basics, were particularly popular in terms of these metrics. Obviously company name and size impact this, but other businesses were in the mix as well. Ultimately, similar products with similar reviews and metrics tend to be grouped together as a result of this metric.

The consistently high ratings (about a 4 out of 5) for these products show that customers generally like dependable, well-regarded products. In addition, the regular emergence of products with significant discount percentages (between 50 and 60 percent) suggests that products with discounts draw in comparable customer segments. What Euclidean similarity displayed is the impact of customer ratings and discounts on the likelihood that customers will buy a similar product. This is obvious, to a degree, but seeing the specific impact it has can help with creating a reasonable price range for selling.